



FISHES of SAHUL

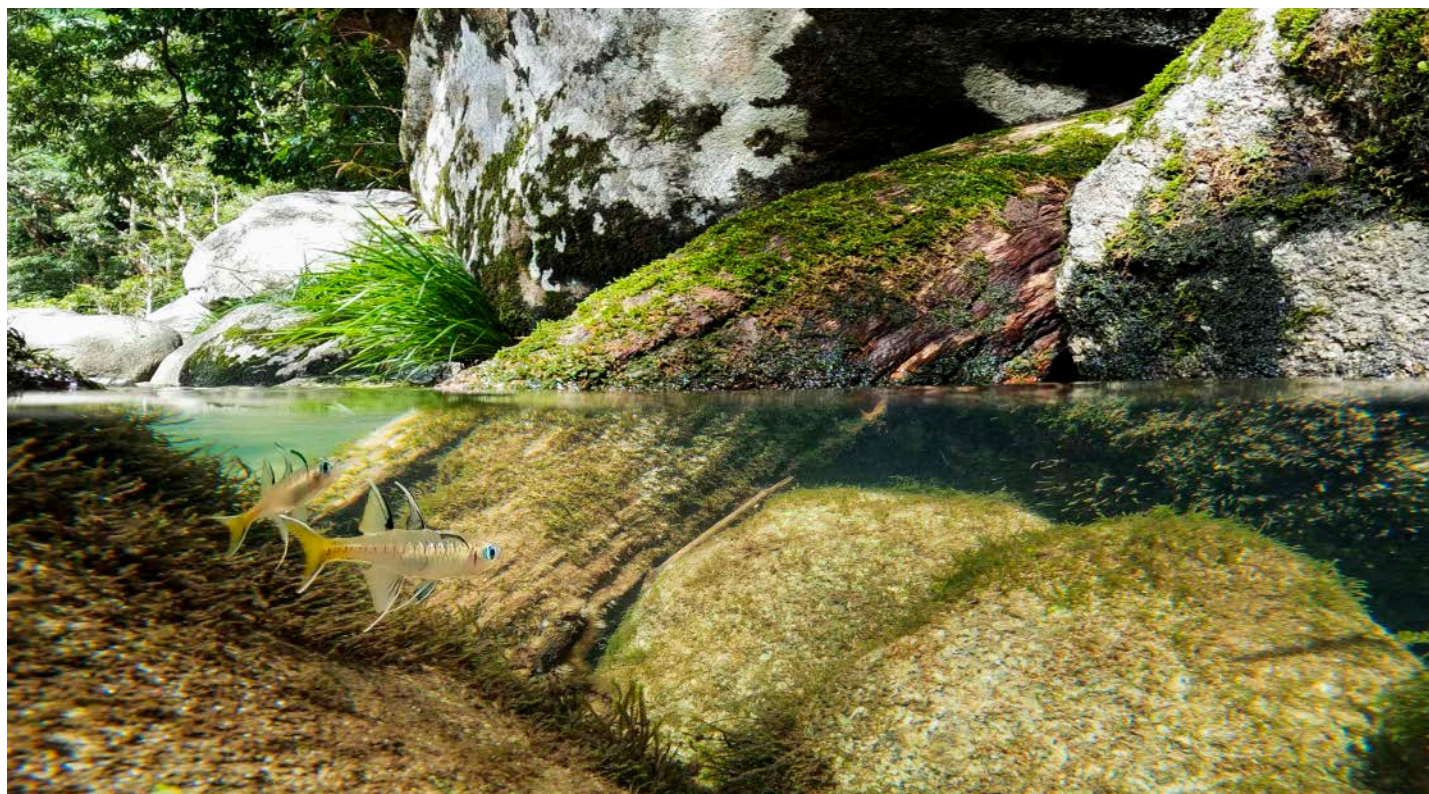


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Pacific Blue-eye *Pseudomugil signifer*, Mossman Gorge, Qld.

Photo: Jason Sulda

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A remarkable radiation of five species in the genus *Kiunga* (Teleostei: Pseudomugilidae) from the Fly River system, Papua New Guinea with descriptions of three new species

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Abstract

The genus *Kiunga* Allen, which is restricted to foothill tributary streams of the Upper Fly River system of Papua New Guinea, is reviewed. The group contains small delicate fishes with a combination of features that include keel-like structures midventrally between the pelvic and anal fins, scale modifications forming rigid sheaths at the bases of the dorsal and anal fins, a microscopic first dorsal fin composed of 1–3 (usually 2–3) spines, segmented dorsal and anal-fin rays 11–19 and 14–20 respectively, lateral scales 28–30, a non-protrusible upper jaw, thorn-like and villiform external teeth on the premaxilla and anterior portion of the dentary, and a semitransparent body with a distinct lack of external pigmentation. The genus contains five species including the previously described *K. ballochi* and *K. bleheri*, and three new taxa described herein. *Kiunga filamentosa*, described from 18 specimens, 18.2–29.6 mm SL, and *K. auromarginata*, described from 8 specimens, 18.2–24.5 mm SL, differ from the other members of the genus in having a higher number of dorsal (15–19 versus usually 12–14) and anal-fin rays (16–20 versus usually 15–16) and less elevated dorsal and anal fins. *Kiunga leucozona*, is described from 51 specimens, 15.1–33.3 mm SL. It is most similar to *K. ballochi*, but differs in usually having 2 dorsal spines (versus 66% with 3 spines in *K. ballochi*) and a white versus yellow submarginal zone on the median fins. The known distribution of this genus lies within an area with a periphery of approximately 265 km, occupying an area of about 3,200 km². The radiation of a monophyletic mini-species flock of this scale within a small section of a single river basin is unknown for any other New Guinea or Australian fish groups.

Introduction

The genus *Kiunga* Allen, 1983 is a member of the family Pseudomugilidae, which contains tiny, delicate fishes that inhabit fresh, brackish, and marine waters of New Guinea and Australia. Family members are commonly referred to as blue-eyes due to the blue colour of the iris that is typical of most species. They were formerly included in the rainbowfish family Melanotaeniidae, but a study by Saeed et al. (1989) provided evidence for separate family recognition, which was further supported by phylogenetic results (Campanella et al., 2015). Blue-eyes are most closely related to sailfin silversides (Telmatherinidae), which are primarily endemic to the island of Sulawesi, Indonesia, with one species, *Kalyptatherina helodes* found in marine environments in western Papua, Indonesia. The family Pseudomugilidae was reviewed by Allen (1995), who recognised 15 species in 3 genera (*Kiunga*, *Pseudomugil* Kner, 1866, and *Scaturiginichthys* Ivantsoff et al., 1991). Three additional species of *Pseudomugil* from New Guinea have been described since that time (Allen et al., 1998; Allen & Renyaan, 1999; Allen et al., 2016).

Although *Kiunga* appears to be closely related and possibly synonymous with *Pseudomugil* (Campanella et al., 2015), Saeed et al. (1989) considered the single species, *K. ballochi* Allen, 1983 that was known at the time of their review as being sufficiently distinct to warrant separate subfamily status. This conclusion was based on a combination of features including a tiny, inconspicuous first dorsal fin, keel-like structure between the pelvic and anal fins, external villiform teeth on the jaws, and several osteological differences. More recent genetic data suggests *Kiunga* is most

closely related to *Pseudomugil connieae* (Allen, 1981) and *P. furcatus* Nichols, 1955 (Unmack et al. unpublished).

Until recently, the genus *Kiunga* included only two species, *K. ballochi* and *K. bleheri* Allen, 2004, both from the general vicinity of Kiunga settlement in the Upper Fly River system of Papua New Guinea (Allen 1983, 2004) and separated by approximately 15–20 km (Fig. 1). However, ongoing surveys in the general area have revealed a wider distribution for the genus (Amick et al., 2020). During oil exploration baseline surveys in 2014, several specimens of what nominally appeared to be *K. ballochi* were collected from three creeks to the east of Kiunga, in the Elevala River subcatchment, and an additional population from a tributary of the Upper Fly River, north of Kiunga. These records extended the known range of *Kiunga* by approximately 40 km, and to the east of the Fly River main channel. Specimens were identified as *K. ballochi* by G. Allen and A. Storey from scanned photographs, but the identification could not be confirmed as no specimens were retained. Additional specimens were collected in October 2014 in a previously unexplored area to the west of the main Ok Tedi (“ok” means river in the local Min language) in small tributaries of the Ok Birim. Andrew Storey and OTML collected specimens of *Kiunga* in October 2018 from one of the sites previously surveyed in the Elevala subcatchment, and from creeks along the newly constructed road to Drimkas, north of Kiunga. These collections show that *Kiunga* is represented by at least three new species, which are described herein.

Materials and Methods

Morphometrics: Counts of fin spines are given in Roman numerals and soft (segmented) rays in Arabic. The last ray of the second dorsal and anal fins is divided near the base and counted as a single element. Gill raker counts refer to the first branchial arch and include the 1–2 rudimentary elements on the upper limb. Other counts and measurements were taken as follows: *lateral scales* - number of scales in horizontal row from upper corner of gill cover to caudal-fin base (hypural junction); *transverse scales* - number of scales in vertical row between anal fin origin and base of dorsal fin; *predorsal scales* - number of scales along midline of nape in front of first dorsal fin; *cheek scales* - total number of scales covering the suborbital and preoperculum; *scales on opercular series* - total number of scales covering the opercle bones; *standard length* (SL) - measured from the tip of the upper lip to the caudal-fin base (hypural junction); *head length* (HL) - measured from the tip of the upper lip to the upper rear edge of the gill opening; *body depth* - the greatest depth at the base of the anal-fin origin; *body width* - measured just posterior to the opercular flap; *snout length* - measured between the front of the upper lip and the anterior edge of the bony orbit; *eye diameter* - horizontal measurement across middle of eye; *interorbital width* - the least bony width; *caudal peduncle depth* - is the least depth; *caudal peduncle length* - measured between two vertical lines, one passing through the base of the last anal ray and the other through the caudal-fin base; *predorsal, preanal, and prepelvic distances* - measured from tip of upper lip to respective origins of first dorsal fin, anal fin, and pelvic fins; *pectoral and pelvic fin lengths* - measured from the base to the tip of the longest ray; *caudal fin length* - measured from the base of the fin (hypural junction) to the tip of the longest ray.

Data in parentheses in the new species descriptions refer to range of values for paratypes if differing from the holotype, and mean value for all specimens. New data were recorded for the previously described species (*Kiunga ballochi* and *K. bleheri*) based on type and non-type specimens. Proportional measurements for the holotype and selected paratypes of the new species are presented as percentage of the standard length in Tables 3–5. Similar tables for *K. ballochi* and *K. bleheri* were included in their original descriptions (Allen 1983 & 2004).

Type specimens are deposited at the Western Australian Museum, Perth (WAM) except for two paratypes of *K. bleheri* at the Australian Museum, Sydney (AMS). Additional specimens included in genetic analyses are deposited at the Museum and Art Gallery of the Northern Territory (NTM).

DNA methods, genomic DNA: We included a total of 33 *Kiunga* individuals from the five species (Table 1). DNA was extracted and sequenced by Diversity Arrays Technologies (DART Pty Ltd, Canberra, Australia, www.diversityarrays.com) based on a combination of complexity reduction using restriction enzymes, implicit fragment size selection, and next generation sequencing (Kilian et al., 2012). All details of the sequencing methods follow Georges et al. (2018). The restriction enzyme combination of PstI (recognition sequence 5'-CTGCA|G-3') and SphI (5'-GCATG|C-3') was used for the double digestion.

The SNP data and associated metadata were read into a genlight object (adegenet, Jombart, 2008) to facilitate processing with package dartR 2.9.4 (Gruber et al., 2018). We first conducted preliminary analyses to identify and remove any individuals that had either excessive missing data due to poor quality DNA,

or excessively high heterozygosity resulting from low levels of tissue/DNA cross-contamination. We then generated two subsets of the SNP data, the “phylo” dataset (starting with 17,620 SNPs for 33 fish) by filtering for reproducibility ≥ 0.99 (DART always undertake repeat sequencing of ~33% of individuals to assess locus reliability) which reduced the number of SNPs to 15,514. Filtering for missing data by locus ≥ 0.5 reduced the number of SNPs to 15,298 with 15.8% missing data overall. A further, more stringent filtering step for missing data by locus = 1 (4,487 SNPs) and removing secondaries using the “best” option reduced the number of SNPs to 4,469 with no missing data to create the “PCoA” dataset for the analyses where the presence of missing data and linked loci are problematic (PCoA and fixed difference calculations).

For the phylo dataset each SNP was concatenated together and analysed with IQ-TREE 2.2.2.7 (Minh et al., 2020) run on the CIPRES server (Miller et al., 2010) incorporating IQ-TREE’s model selection procedure (-m TEST; Kalyanamoorthy et al., 2017), resulting in the selection of the TVM+F model followed by 10,000 replicates of ultrafast bootstrapping with the SH-like approximate likelihood-ratio test (-aLRT 1000; Hoang et al., 2018). Trees were mid-point rooted due to the lack of suitable outgroups. Ordination using the PCoA dataset via principle components analysis (PCoA, Gower, 1966), with individuals as entities and loci as attributes as implemented by the `gl.pcoa` and `gl.pcoa.plot` functions of dartR. Fixed differences were calculated using the `gl.fixed.diff` function in dartR.

DNA methods, mitochondrial DNA: We included a total of 20 *Kiunga* individuals from the five species (Table 1). We sequenced the first 601 base pairs of the Cytochrome *b* gene using the primers Glu18 (TAACCAGGACTAATGRCTTGAA) and Hd.alt (GGRTTGTGGAGCCTGTTTCAT) following the methods given in Allen & Unmack (2012). Sequence data were analysed in the same manner as the SNPs using the HKY+F+G4 model. All sequences obtained in this study were deposited in GenBank, accession numbers PQ128253–PQ128272.

Kiunga Allen, 1983

Kiunga Allen, 1983: 72 (type species: *Kiunga ballochi* Allen, 1983 by original designation; treated as feminine).

Diagnosis: A member of the family Pseudomugilidae, but differing from other genera by a combination of features that include: keel-like structures midventrally between pelvic and anal fins; scale modifications forming rigid sheaths at bases of dorsal and anal fins; microscopic first dorsal fin composed of 1–3 (usually 2–3) spines; segmented dorsal and anal-fin rays 11–19 and 14–20, respectively; lateral scales 28–30; transverse scales 5–6; semitransparent body with distinct lack of external pigmentation and silvery peritoneum, vertebrae, ribs, and pterygiophores clearly evident; upper jaw non-protrusible; thorny to villiform external teeth on premaxilla and anterior portion of dentary.

Distribution and habitat: The geographic distribution of the five species in the genus is remarkable, being restricted to a very limited area within the greater Fly River system (Fig. 1). The known distribution lies within an area with a periphery of approximately 265 km, occupying an area of about 3,200 km². The radiation of a monophyletic mini-species flock of this scale within a single river basin is unknown for any other New Guinea or Australian fish

Table 1. Samples and locality details of specimens used in genetic analyses.

Species	Site	Latitude	Longitude	Date	DNA code	SNPs	Cytb
<i>Kiunga auromarginata</i>	Taki Creek, Elevala	-6.1400	141.7427	15/10/2018	KbTaki	5	2
<i>Kiunga ballochi</i>	Kiunga Hwy, 42.5 km from Kiunga	-5.8655	141.1836	18/08/2007	Kball	2	2
<i>Kiunga ballochi</i>	Bige Hwy, 13.8 km from Bige Ops	-5.8709	141.1743	2011	KBBi13.8	4	-
<i>Kiunga bleheri</i>	Konkonda Hwy, 10.0 km from Kiunga turnoff	-6.0630	141.2360	15/08/2007	Kbleh	2	2
<i>Kiunga bleheri</i>	Konkonda Hwy, km 10.0 marker	-6.0630	141.2360	2011	KBKonkonda9	4	-
<i>Kiunga filamentosa</i>	Drimkas Hwy, site DRIM-3	-5.9221	141.3925	14/10/2018	KbDrimk	6	3
<i>Kiunga leucozona</i>	Ok Indok	-5.7045	141.0759	16/10/2014	KBBi10	2	2
<i>Kiunga leucozona</i>	Ok Yep	-5.6986	141.0484	19/10/2014	KBBi17	2	2
<i>Kiunga leucozona</i>	Ok Yep	-5.6866	141.0338	19/10/2014	KBBi18	1	2
<i>Kiunga leucozona</i>	Ok Yep	-5.7067	141.0236	20/10/2014	KBBi20	1	2
<i>Kiunga leucozona</i>	Ok Bim	-5.7066	141.0296	20/10/2014	KBBi21	2	2
<i>Kiunga leucozona</i>	Ok Teelleck	-5.7081	141.0379	20/10/2014	KBBi22	2	1

Table 2. Comparison of fin-ray counts, number of cheek (preopercle) scales, and mean maximum body depth (as % of SL) for species of *Kiunga*. Pectoral rays were counted on both sides of each specimen.

Dorsal rays																			Anal rays					
Species	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29					
<i>K. ballochi</i>			21	15	2								7	23	8									
<i>K. bleheri</i>	2	10	5										6	6	4	1								
<i>K. auromarginata</i>						2	5	1								1	1	3	3					
<i>K. filamentosa</i>					9	7	1	1								5	7	6						
<i>K. leucozona</i>		19	29	3						6	15	28	2											
Pectoral rays																			Mean body depth					
Species	10	11	12	13			1	2	3	4	5	No.												
<i>K. ballochi</i>		17	47					17	21			30.3	15											
<i>K. bleheri</i>	1	20	4				4	10				34.0	12											
<i>K. auromarginata</i>		10	4	1					4	3	1	26.0	8											
<i>K. filamentosa</i>		33	3						10	8		29.0	15											
<i>K. leucozona</i>	3	62	35	1			1	47	3			31.7	20											

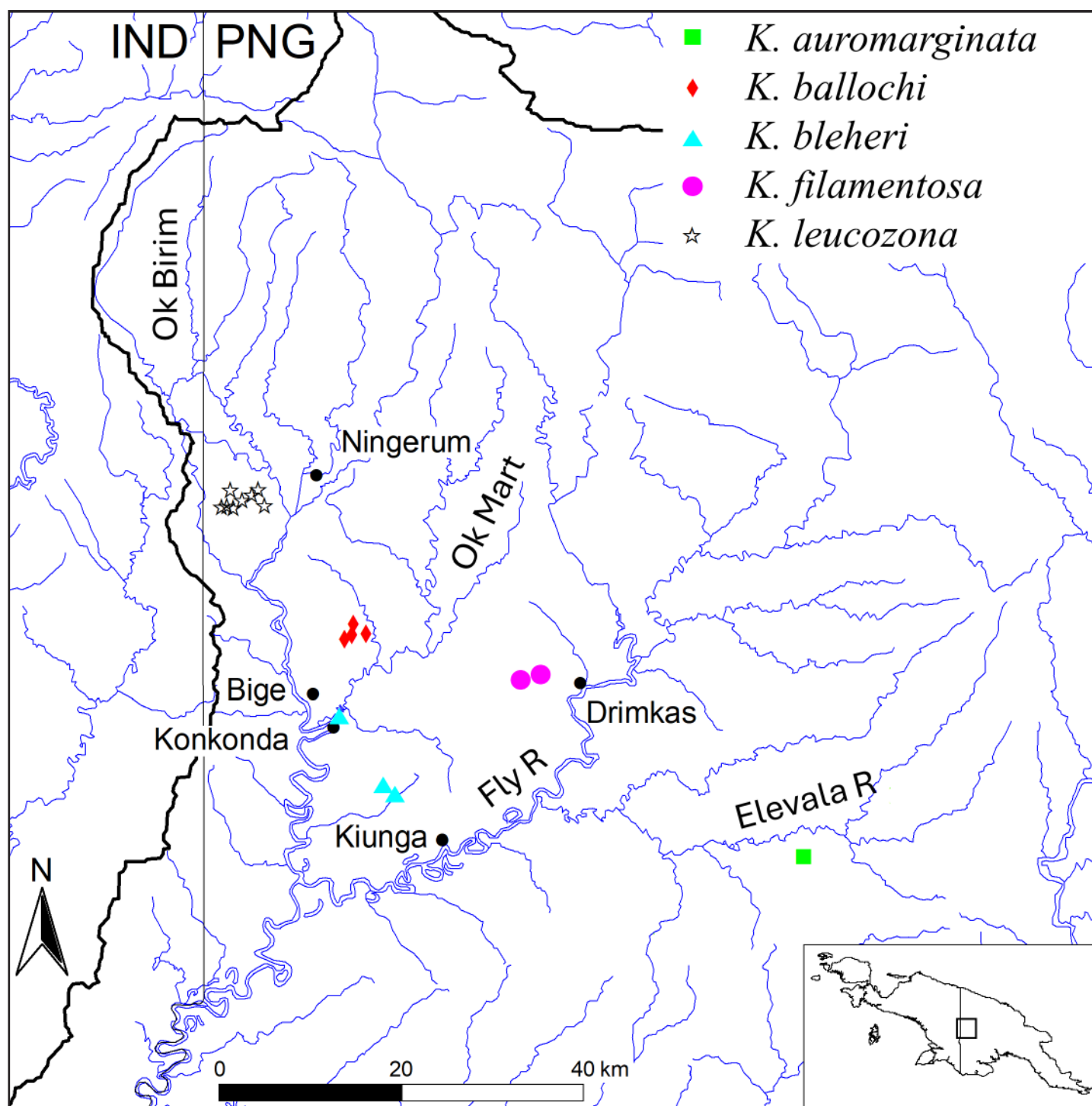


Figure 1. Distribution maps for species of *Kiunga*.

groups. The only other similar radiations from limited geographic areas involve either a lacustrine environment (Lake Kutubu, Papua New Guinea with seven species of *Mogurnda*, family Eleotridae) or a fragmented insular setting (Aru Islands with seven non-monophyletic species of *Melanotaenia*, family Melanotaeniidae).

The typical habitat consists of clear, shallow streams in relatively flat or low foothill terrain amongst undisturbed rainforest at elevations of about 40–100 m above sea level. Streams are relatively well shaded, but have occasional open patches exposed to sunlight.

Remarks: In addition to the features listed in the above diagnosis, Saeed et al. (1989) mentioned several osteological features that were

apparently different than those found in other pseudomugilids, including: first element of second dorsal fin supported by two pterygiophores, with first pterygiophore almost reaching pterygiophore of first dorsal fin; maxilla articulating with anterior arm of lateral ethmoid via small submaxillary meniscus; maxillary condyle on vomer absent; metapterygoid absent. However, during the present study we have not determined if these features are consistent for the entire genus. The group is known to local people, with the name Wasum provisionally recorded for Faiwol and bordering Awin languages (Micah Borok pers. comm. 2024), and further work will likely document language names [note: a Suki name for *K. ballochi* listed in Allen et al., (2008) likely applies to a different fish from the mid-Fly].

Key to the species of *Kiunga*

- 1a.** Second dorsal and anal fins relatively short with subequal rays, longest 10.9–19.9 (mean 14.7)% SL; segmented rays of dorsal and anal fins usually unbranched; anal-fin spine nearly as long as first segmented anal-fin ray; six horizontal scale rows between anal-fin origin and base of second dorsal fin.....*K. bleheri*
- 1b.** Second dorsal and anal fins relatively tall with longest rays anteriorly, longest 18.1–35.9 (mean 19.4–27.6)% SL; segmented rays of dorsal and anal fins usually branched; anal-fin spine much shorter than first segmented anal-fin ray; five horizontal scale rows between anal-fin origin and base of second dorsal fin..... **2**
- 2a.** Segmented dorsal-fin rays 15–19; segmented anal-fin rays 16–20, usually 16 or more; second dorsal and anal fins moderately elevated anteriorly, mean length of longest dorsal ray 20.8–21.6% SL and mean length of longest anal ray 19.4–22.1% SL..... **3**
- 2b.** Segmented dorsal-fin rays usually 12–14 (infrequently 11 or 15); segmented anal-fin rays 14–17, usually 15–16 (except 22% of *K. ballochi* with 17); second dorsal and anal fins greatly elevated anteriorly, mean length of longest dorsal ray 26.6–27.2% SL and mean length of longest anal ray 26.7–27.6% SL..... **4**
- 3a.** Body relatively slender, greatest depth 25.5–27.0, mean 26.0% of SL; pelvic fins short and non-filamentous, generally not reaching anal-fin origin when adpressed.....*K. auromarginata*, **n. sp.**
- 3b.** Body deeper, greatest depth 27.8–31.4, mean 29.0% of SL; pelvic fins of presumed males long and filamentous if undamaged, reaching well beyond anal-fin origin when adpressed.....*K. filamentosa*, **n. sp.**
- 4a.** Dorsal spines 2–3, most frequently 3 (66%); first dorsal spine length 2.4–6.0 in HL; cheeks scales 2–3 (55% with 3); live individuals with prominent submarginal yellow zone on median fins; distributed east of Ok Tedi.....*K. ballochi*
- 4b.** Dorsal spines 1–3, infrequently 3 (18%); first dorsal spine length 1.5–3.9 in HL; cheeks scales 1–3 (92% with 2); live individuals with prominent submarginal white zone on median fins; distributed west of Ok Tedi.....*K. leucozona*, **n. sp.**

Kiunga ballochi Allen, 1983

(Figs. 2–3; Table 2)

Kiunga ballochi Allen, 1983: 73 (type locality: small tributary of Upper Fly River about 40 km north of Kiunga on Tabubil Road, Papua New Guinea).

Material examined: All from small tributaries of the Ok Tedi system along the Kiunga-Tabubil Road, approximately 8–12 km northwest of Rumginae Village. WAM P.27797-001 (holotype), 23.5 mm SL; WAM P.27797-002 (paratypes), 35 specimens, 14.7–23.1 mm SL; WAM P.27798-001 (paratypes), 21 specimens (paratypes), 17.4–27.3 mm SL.

Diagnosis: Dorsal-fin elements II–III + 13–15 (usually III + 13–14, mean 13.5); anal-fin elements I,15–17, mean 16.0; second dorsal and anal fins relatively tall with triangular profile, longest dorsal ray 21.1–28.8, mean 26.6% SL and longest anal ray 21.3–31.5, mean 27.6% SL; first dorsal spine 2.4–6.0, mean 4.0% SL; anal spine relatively short, 5.9–8.1, mean 7.3% SL; lateral scales 28; transverse scales 5; cheek scales 2–3 (55% with 3); predorsal scales 12–14 (usually 13), including pair of enlarged scales anteriorly; greatest body depth 28.7–32.9, mean 30.3% of SL; caudal fin forked; colour in life mainly semitransparent with silvery to blackish midlateral stripe and silvery peritoneum clearly evident; second dorsal, anal, and caudal fins with relatively narrow black outer margin and adjacent, much wider submarginal yellow zone, also yellow suffusion on membranes of dorsal and anal fins; pelvic fins yellow; maximum known size 29.2 mm SL.

Redescription: Based on holotype and 38 paratypes, 21.0–29.2 mm SL. Dorsal-fin elements II–III + 13–15 (usually III + 13–14, mean

13.5); anal-fin elements I,15–17, mean 16.0; pectoral-fin rays 11–12 (73% with 12); pelvic-fin elements I,5; lateral scales 28; transverse scales 5; scales between dorsal fins 3; predorsal scales 12–14 (usually 13), including two enlarged scales anteriorly; cheek scales 2–3 (55% with 3); scales on opercular series 5; gill rakers on lower limb of first arch 7–8.

Body depth 3.0–3.5, 3.3, head length 3.7–4.4, 4.0, pre-first-dorsal distance 2.0–2.2, 2.1, pre-second-dorsal distance 1.6–1.8, 1.7, preanal distance 1.7–2.0, 1.8, prepelvic distance 2.5–2.8, 2.6, all in SL. Greatest width of body 2.0–2.7, 2.2 in body depth. Snout length 3.1–4.5, 3.8, eye diameter 2.0–2.5, 2.3; interorbital width 2.3–3.0, 2.7; depth of caudal peduncle 2.1–2.4, 2.3, length of caudal peduncle 1.1–1.3, 1.2, all in head length.

First dorsal fin originates about one-fourth to one-half eye diameter posterior to level of anal fin origin; first dorsal spine 4.1–9.5, 6.5, longest ray of second dorsal fin 0.9–1.2, 0.9, of anal fin 0.8–1.2, 0.9, all in head length. Most of soft (segmented) rays of second dorsal fin branched except first ray frequently unbranched in specimens less than about 20 mm SL. Pelvic fins short, the tips not reaching origin of anal fin when adpressed; innermost pelvic fin ray connected to abdomen along most of length by delicate membrane (usually damaged during collecting or examination); urogenital openings positioned on mid-ventral line about one-third to one-half distance between pelvic fin origin and posterior tip of depressed pelvic fins; length of pelvic fin 1.7–2.1, 1.9 in head length. Pectoral fins pointed, longest rays 1.2–1.6, 1.4 in head length. Caudal fin emarginate, its length 3.5–4.1, 3.8 in SL.

Jaws oblique, mouth terminal; maxilla extends posteriorly to level between anterior edge of eye and anterior edge of pupil. Upper



Figure 2. *Kiunga ballochi*, freshly captured specimen in aquarium, approximately 24 mm SL, Rumginae area, Papua New Guinea.

Photo: Mark Allen



Figure 3. *Kiunga ballochi*, preserved holotype, 23.5 mm SL, 30 km north-west of Kiunga, Papua New Guinea.

Photo: Mark Allen

jaw (premaxilla) covered with about 45 to 60 teeth arranged in 2 to 4 irregular rows on each side, mostly exposed when mouth closed, most teeth small and villiform with sharp tips, largest teeth situated along inner edge of rear portion of premaxillary, including several forward slanting teeth that are larger than others. Lower jaw with less conspicuous, small, backward slanting villiform teeth confined mainly to anterior portion, about 25 to 30 on each side of median symphysis.

Scales cycloid, relatively large and elongated anteroventrally, arranged in regular horizontal rows; predorsal scales extending to middle of interorbital, including pair of relatively enlarged scales on frontal-interorbital; 2–3 scales on cheek just below eye, those with 2 scales likely missing one scale; pores of cephalic sensory pores relatively large and conspicuous, 9 in preopercle series and 10 in supraorbital series; nostril openings relatively small, anterior naris placed on side of snout next to premaxillary and posterior naris situated next to upper anterior corner of eye.

Colour in life (Fig. 2): mainly semitransparent silvery peritoneum clearly evident; opercle with silvery to pinkish (due to underlying gills); iris silvery bluish, darker on dorsal and ventral portions; thin blackish stripe along middle of side flanked by wider, internal yellow stripes; scales on upper half of body thinly outlined with pepper-like melanophores; mid-dorsal line between origin of second dorsal fin and interorbital, ventral edge of caudal peduncle, and bases of dorsal, caudal, and anal fins with concentrations of melanophores, most strongly developed across base of caudal fin, also mid-dorsal and mid-ventral profile frequently with yellow suffusion; tiny first dorsal fin translucent yellow; second dorsal, caudal, and anal fins with bold black margins and adjacent, much wider yellow zone, remaining portions of these fins transparent to slightly yellowish except dorsal and anal fins with narrow, bright yellow stripe on basal-most portion; pelvic fins yellow; pectoral fins translucent; all segmented fin rays faintly outlined with black.

Colour in alcohol (Fig. 3): overall yellowish white with scale margins on upper half of body narrowly outlined with black and thin black stripe on middle of side from pectoral region to base of caudal fin; concentration of melanophores on mid-dorsal line between origin of second dorsal fin and interorbital and similar concentration of dark pigment on dorsal and ventral midlines of caudal peduncle and along bases of second dorsal and anal fins; fins mainly semi-translucent, outer edges of second dorsal, anal, and caudal fins with prominent black borders.

Comparisons: Genetic and morphological analysis reveal that *K. ballochi* is most closely related to *K. leucozona*. These species differ significantly from the highly divergent *K. bleheri* by their different shaped dorsal and anal fins (roughly triangular versus rectangular), forked versus emarginate caudal fin, five versus six transverse scales, 28 versus 30 lateral scales, and mainly branched versus unbranched segmented dorsal and anal-fin rays. They differ from *K. auromarginata* in possessing a deeper body (mean depth 30.3–31.7 versus 26.0% SL) and from both *K. auromarginata* and *K. filamentosa* in having fewer segmented dorsal-fin rays (usually 12–14 versus 15–19). Both species also lack filamentous pelvic fins that sometimes extend beyond the anal-fin origin in *K. filamentosa*, presumably a feature restricted to adult males. Other than the genetic difference (see separate section at end of article) the closely related *K. ballochi* and *K. leucozona* are separable on the basis of colour pattern, notably the presence of yellow submarginal zones on the median fins of *K. ballochi* compared to the white zones on the fins of *K. leucozona*. There are also strong modal differences in the number of dorsal spines and cheek scales. Although both species possess overlapping counts, *K. ballochi* most frequently (66% of specimens) has three dorsal spines compared with only 18% with three spines in *K. leucozona*, which usually has two spines. Similarly, *K. ballochi* frequently has three cheek scales (55% of specimens) compared to the preponderance (92% of specimens) of two cheek scales in *K. leucozona*.

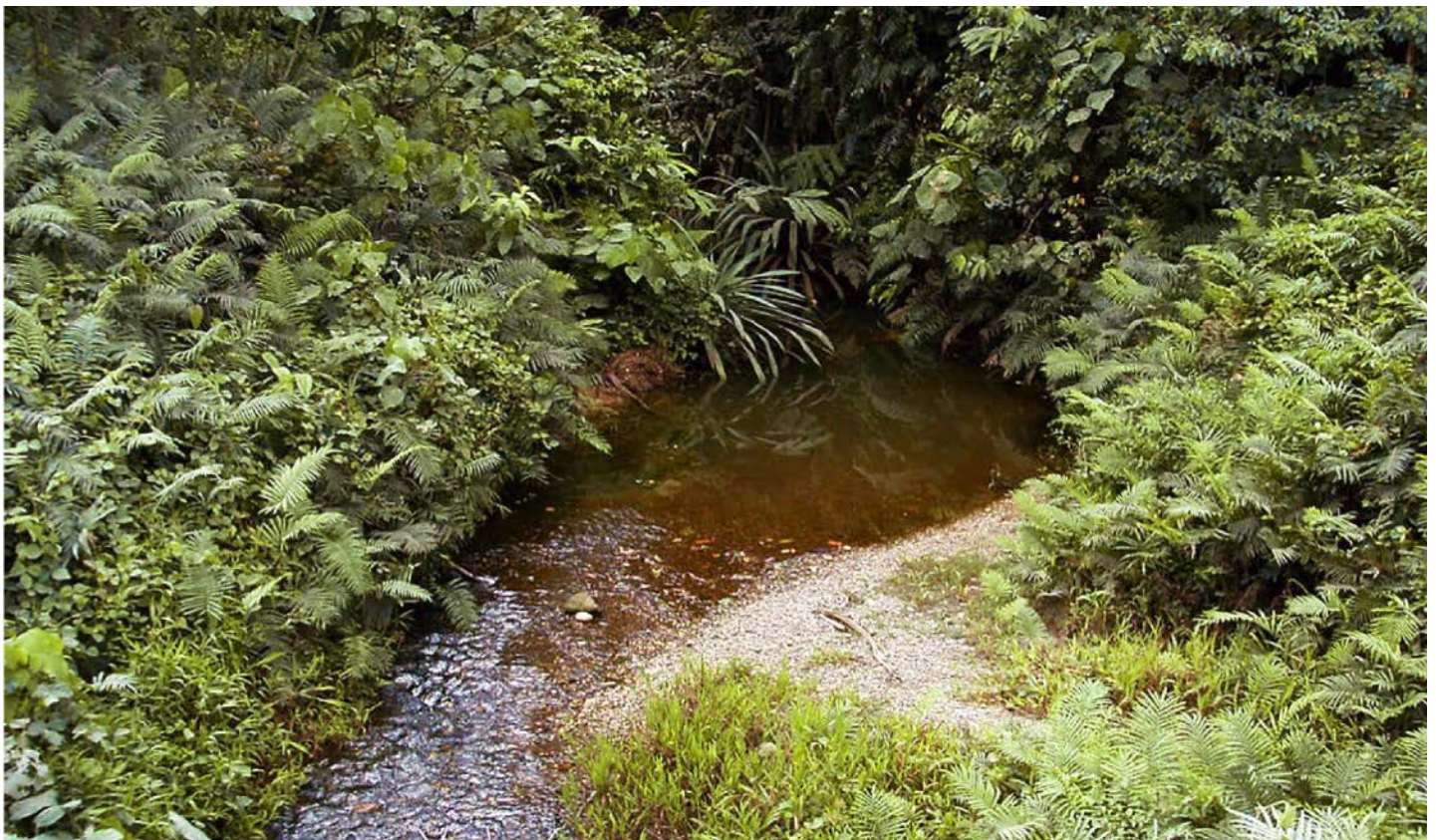


Figure 4. Creek at KM 42.5 on Kiunga-Tabubil Highway where *Kiunga ballochi* was collected in 2007.

Photo: Andrew Storey



Figure 5. *Kiunga bleheri*, freshly captured specimen in aquarium, approximately 25 mm SL, Konkonda Road, Kiunga district, Papua New Guinea. Photo: Mark Allen

Distribution and habitat: This species is currently known from several small creeks situated along or close to the Kiunga-Tabubil Road (Fig. 1). The area of distribution covers approximately 20 km² and is centered on a point about 10 km north of Rumginae Village. The habitat (Fig. 4) consists of clear, shallow creeks flowing through dense rainforest with occasional sunlit clearings. A temperature of 24.5°C and a pH of 7.8 were recorded at the type locality. It occurred in relatively large aggregations when first collected in 1982, but only small numbers were observed when it was rediscovered in 2007 (Allen, 2007).

Etymology: The species was named in honour of David Balloch, a former senior biologist with Ok Tedi Mining Limited, who facilitated the collection of the type specimens.

Kiunga bleheri Allen, 2004

(Figs. 5–6; Table 2)

Kiunga bleheri Allen, 2004: 80 (type locality: Tare Creek, 11.5 kilometers from central market of Kiunga on Konkonda Road, about 6°04'S, 141°13'E, Western Province, Papua New Guinea).

Material examined: All specimens from Konkonda Road, Kiunga district, Western Province, Papua New Guinea. AMS I.42990-001 (paratypes), 2 specimens, 24.1–24.4 mm SL; WAM P.32455-001 (holotype), 28.4 mm SL; WAM P.32455-002 (paratypes), 4 specimens, 16.7–25.8 mm SL; WAM P.31572-001 (paratypes), 2 specimens, 20.0–25.4 mm SL; WAM P.34003-001, 8 specimens, 17.1–26.1 mm SL.

Diagnosis: Dorsal-fin elements II–IV + 15–18 (usually II + 11–13, mean 12.2); anal-fin elements I, 15–18, mean 16.0; second dorsal and anal fins relatively low with most rays subequal, longest dorsal ray 10.9–19.9, mean 14.7% SL and longest anal ray 11.6–19.2, mean 15.3% SL; first dorsal spine 4.3–7.0, mean 5.5% SL; anal spine relatively long, 8.7–12.4, mean 10.6% SL; lateral scales 30; transverse scales 6; cheek scales 1–2 (usually 2); predorsal scales 14–15, including pair of enlarged scales anteriorly; greatest body depth 29.7–36.6, mean 34.0% of SL; caudal fin slightly emarginate; colour in life mainly semitransparent with narrow dark midlateral stripe, silvery peritoneum clearly evident and adults with outer profile of anterior portion of body with yellowish to orange hue, encompassing predorsal, interorbital, snout, jaws, lower head, and prepelvic midline; fins semitransparent with narrow black margins on second dorsal, anal, and pelvic fins; caudal fin usually plain, but occasionally with diffuse blackish pigment along posterior margin maximum known size 28.4 mm SL.

Redescription: Based on holotype and 11 paratypes, 19.1–28.4 mm SL. Dorsal-fin elements II–IV + 15–18 (usually II + 11–13, mean 12.2); anal-fin elements I, 15–18, mean 16.0; pectoral-fin rays 10–12 (usually 11); pelvic-fin elements I, 5; lateral scales 30; transverse scales 6; scales between dorsal fins 3; predorsal scales 14–15 (usually 14), including two enlarged scales anteriorly; preopercle scales 1–2 (usually 2); scales on opercular series 5; gill rakers on lower limb of first arch 8.

Body depth 2.7–3.4, 3.0; head length 3.7–4.5, 3.0, pre-first-dorsal distance 1.8–2.1, 1.9, pre-second-dorsal distance 1.5–1.6, 1.6, preanal distance 1.8–2.2, 1.9, prepelvic distance 2.5–2.7, 2.6, all in SL. Greatest width of body 1.8–2.5, 2.3 in body depth. Snout



Figure 6. *Kiunga bleheri*, preserved specimen (WAM P.34003-001), 25.8 mm SL, Konkonda Road, Kiunga district, Papua New Guinea.

Photo: Gerald Allen

length 3.4–4.4, 3.9, eye diameter 2.1–2.5, 2.3; interorbital width 2.2–2.6, 2.4; depth of caudal peduncle 1.9–2.3, 2.1, length of caudal peduncle 1.1–1.4, 1.3, all in HL.

First dorsal fin originates about opposite of anal fin origin; first dorsal spine 3.4–6.0, 4.5, longest ray of second dorsal fin 1.3–2.0, 1.7, of anal fin 1.3–2.1, 1.6, all in HL. Most of soft (segmented) rays of second dorsal fin unbranched except last 1–2 rays frequently branched. Last 1–2 segmented rays of anal fin frequently branched, except last 8 rays branched in holotype. Pelvic fins short, the tips not quite reaching origin of anal fin when depressed; innermost pelvic fin ray connected to abdomen along most of length by delicate membrane (usually damaged during collecting or examination); urogenital openings positioned on mid-ventral line about one-third to one-half distance between pelvic fin origin and posterior tip of depressed pelvic fins; length of pelvic fin 1.5–2.2, 1.8 in HL. Pectoral fins pointed, longest rays 1.2–1.6, 1.4 in HL. Caudal fin emarginate, its length 3.8–5.5, 4.6 in SL.

Jaws oblique, mouth terminal; maxilla extends posteriorly to level between anterior edge of eye and anterior edge of pupil. Upper jaw (premaxilla) covered with about 46 to 60 teeth arranged in 2 to 4 irregular rows on each side that are exposed when the mouth is closed, most teeth small and villiform with sharp tips, largest teeth situated along inner edge of rear portion of premaxillary, including several forward slanting teeth that are larger than others. Lower jaw with less conspicuous, small, backward slanting villiform teeth confined mainly to anterior portion, about 15 to 20 on each side of median symphysis.

Scales cycloid, relatively large and elongated anteroventrally, arranged in regular horizontal rows; predorsal scales extending to middle of interorbital, including 2–3 relatively enlarged scales on frontal-interorbital; 4 (3–4) scales on cheek just below eye, those with 3 scales likely missing one scale; pores of cephalic sensory pores relatively large and conspicuous, 9 in preopercle series and 7–8 in supraorbital series; nostril openings relatively small, anterior naris placed on side of snout next to premaxillary and posterior naris situated next to upper anterior corner of eye.

Colour in life (Fig. 5): mainly semitransparent with silvery peritoneum clearly evident; and opercle pinkish due to underlying gills; iris golden to bluish; prominent dark brown or blackish stripe along middle of side and thinner black stripe on mid-dorsal line between origin of first dorsal fin and interorbital; adults with outer profile of anterior portion of body with yellowish to orange hue, encompassing predorsal, interorbital, snout, jaws, lower head, and prepelvic midline; uppermost portion of back and upper edge of caudal peduncle sometimes with orange hue; concentrations of pepper-like melanophores along bases of dorsal and anal fin; fins generally semitransparent, except tiny first dorsal sometimes yellowish; second dorsal, anal, and pelvic fins with narrow blackish margin; outer edge of second dorsal fin rarely with submarginal yellowish stripe; caudal fin usually colourless, but occasionally with diffuse blackish pigment along posterior margin; pectoral fin semitransparent.

Colour in alcohol (Fig. 6): overall yellowish white with narrow black stripe on mid-lateral side of body and concentrations of pepper-like melanophores on scale margins of mid-lateral row, upper back, and row just above anal-fin base, also on breast region; fins mainly semi-translucent grey except narrow black margin on dorsal and anal fins, and occasionally on rear edge of caudal fin; less conspicuous black stripe along base of dorsal and anal fins, and blackish bar on caudal-fin base.

Comparisons: *Kiunga bleheri* clearly differs from other members of the genus by its much lower second dorsal and anal fins, which also have a different shape, characterized by most elements being subequal in length. It also differs in having an emarginate rather than forked caudal fin, which usually lacks blackish margins. Other diagnostic features include 30 lateral scale rows, six transverse rows, a relatively long anal-fin spine, which is nearly as long as the first segmented anal ray, and mainly unbranched dorsal and anal fin rays. Genetic analysis (see separate section at end of article) indicates that *K. bleheri* is the most divergent member of the genus.



Figure 7. Tare Creek, type locality of *Kiunga bleheri* on Kiunga-Konkonda Highway, being sampled in August 2005.

Photo: Andrew Storey

Distribution and habitat: This species is currently only known from three small creeks (Figure 1), including the type locality (Tare Creek; Fig. 7), in the immediate vicinity of Konkonda Road, which runs in a northwesterly direction for 22 km from the town of Kiunga to the settlement of Konkonda on the Ok Tedi. The species was first obtained by aquarium-fish collector Heiko Bleher in 1991, who mistakenly identified it as *K. ballochi*. Subsequent observations and collections were made in 2003, 2007, and 2011 (Amick *et al.* 2020).

The type specimens from Tare Creek were collected about 1.5 km upstream from the road crossing. At this point the stream is about 2.5 m in width and during low water periods is only 50–80 cm deep. A group of approximately 80 *K. bleheri* was observed swimming in the middle of the stream at the type locality at the time of collection, where the current was relatively fast. Temperature, pH and conductivity values of 26.2°C, 6.0, and 10 µS/cm were recorded. There was no aquatic vegetation, although the Pandanus-lined stream was littered with a large amount of driftwood and fallen tree debris. The only other fish inhabitants included gudgeons (*Mogurnda* and *Giuris* sp.) and the rainbowfish *Melanotaenia novaeguineae*.

Etymology: The species was named in honour of Heiko Bleher of Milan, Italy who collected the type specimens and was the first person to document it to western science.

***Kiunga auromarginata* n. sp.**

(Figs. 1, 8–9; Tables 2–3)

Holotype: WAM P.35288-001, 24.5 mm SL, Taki Creek, tributary of Elevala River, 4.3 km south of Gasuke Village, 6° 8.404'S, 141° 44.565'E, seine net, A.W. Storey, 15 October 2018.

Paratypes: WAM P. 35288-002, 7 specimens, 18.2–23.7 mm SL, collected with holotype.

Diagnosis: Dorsal-fin elements II + 17–19 (usually II + 18, mean 17.9); anal-fin elements I,17–20, mean 19.0; second dorsal and anal fins moderately tall with triangular profile, longest dorsal ray 18.1–24.5, mean 20.8% SL and longest anal ray 17.1–22.1, mean 19.4% SL; first dorsal spine 2.3–4.0, mean 3.1% SL; anal spine relatively short, 7.4–9.4, mean 8.2% SL; lateral scales 28; transverse scales 5; cheek scales 3–5 (usually 3–4); predorsal scales 12–13, including pair of enlarged scales anteriorly; greatest body depth 25.5–27.0, mean 26.0% of SL; caudal fin forked; colour when fresh mainly semitransparent with faint dark midlateral stripe and silvery peritoneum clearly evident; prominent orange markings include those on jaws/snout tip, tiny first dorsal fin, and outer margins and bases of second dorsal and anal fins; caudal fin with blackish posterior margin and pelvic fins yellowish with orange anterior margin and posterior tips; maximum known size 24.5 mm SL.

Description: Dorsal-fin elements III-13 (I–III + 12–14); anal-fin elements I,16 (I,14–17); pectoral-fin rays 12 (10–13); pelvic-fin elements I,5; branched caudal rays 10; principal caudal rays 12; accessory caudal rays 7 + 7 (6–8 + 6–8); lateral scales 28; transverse scales 5; scales between dorsal fins 3; predorsal scales 12 (12–13); preopercle scales 3 (3–5); scales on opercular series 5; gill rakers on lower limb of first arch 8.

Body depth 3.7 (3.8–3.9, 3.8); head length 4.3 (4.0–4.4, 4.2), pre-first-dorsal distance 2.3 (2.1–2.4, 2.3), pre-second-dorsal distance 1.9 (1.8–2.0, 1.9), preanal distance 1.9 (1.8–2.0, 1.9), prepelvic distance 3.1 (2.7–3.2, 2.9), all in SL. Greatest width of body 2.1 (1.7–2.0, 1.9) in body depth. Snout length 4.8 (3.7–5.1, 4.4); eye diameter 2.2 (2.0–2.1, 2.0); interorbital width 2.5 (2.2–2.8, 2.6); depth of caudal peduncle 2.4 (2.1–2.5, 2.4); length of caudal peduncle 1.2 (1.3–1.4, 1.3), all in HL.

Second dorsal fin originates slightly behind anal fin origin; first dorsal fin spine 10.3 (5.9–9.8, 7.8), longest ray of second dorsal fin 1.0 (1.0–1.3, 1.1), of anal fin 1.1 (1.1–1.4, 1.2), all in HL. All



Figure 8. *Kiunga auromarginata*, freshly captured specimen, approximately 24 mm SL, tributary of Elevala River, Papua New Guinea.

Photo: Andrew Storey

segmented rays of second dorsal and anal fins branched, except first dorsal ray. Pelvic fin tips extending slightly past anal-fin origin in holotype, but well short of anal-fin origin in paratypes, pelvic length 1.1 (1.3–1.6, 1.4) in HL. Pectoral fins pointed, longest rays 1.5 (1.2–1.5, 1.3) in HL. Caudal fin forked, its length 3.9 (3.6–5.0, 4.1) in SL.

Jaws oblique, mouth terminal; maxilla extends posteriorly to about level of anterior edge of pupil. Upper jaw (premaxilla) covered with about 33–35 teeth arranged mainly in two irregular rows on each side, including many that are exposed when the mouth is closed, most teeth small and villiform with sharp tips, largest teeth situated along inner edge of rear portion of premaxillary, including several

forward-slanting teeth that are larger than others. Lower jaw with less conspicuous, small, backward slanting villiform teeth confined mainly to anterior portion, about 10–15 on each side of median symphysis.

Scales cycloid, relatively large and elongated anteroventrally, arranged in regular horizontal rows; predorsal scales extending to middle of interorbital, including pair of enlarged scales on frontal-interorbital; 2 scales on cheek just below eye; pores of cephalic sensory pores relatively large and conspicuous, 8 in dentary/preopercle series and equal number in supraorbital series; nostril openings relatively small, anterior naris placed on side of snout next to premaxillary and posterior naris situated next to upper anterior corner of eye.



Figure 9. *Kiunga auromarginata*, preserved holotype (WAM P.35288-001), 24.5 mm SL, tributary of Elevala River, Papua New Guinea

Photo: Gerald Allen



Figure 10. Taki Creek habitat of *Kiunga auromarginata*. Photo: Andrew Storey

Colour when fresh (Fig. 8): overall semi-translucent whitish with silvery reflections on rear portion of head and adjacent anterior body; prominent orange markings include those on jaws/snout tip, tiny first dorsal fin, and outer margins and bases of second dorsal and anal fins, both fins also possess fine blackish basal stripes and blackish outer edges (sometimes partially blackish); caudal fin with diffuse blackish dorsal and ventral edges and more conspicuous black rear margin with adjacent submarginal band of

pale orange; pelvic fins yellowish with orange anterior margin and posterior tips.

Colour in alcohol (Fig. 9): overall yellowish white with inconspicuous pepper-like melanophores on scale margins, mainly on anterior half of body; single row of close-set melanophores forming midlateral black stripe extending from just above pectoral-fin base to rear portion of caudal

Table 3. Proportional measurements of holotype and paratypes (WAM P.35288-001 and P.35288-002 respectively) of *Kiunga auromarginata* expressed as percentage of the standard length.

	Holotype			Paratypes				
Standard length	24.5	23.7	23.3	20.8	19.8	19.4	19.2	18.2
Head length	23.4	22.6	23.5	24.0	23.5	23.1	24.0	24.9
Body depth	27.0	26.3	25.5	26.0	25.5	26.3	25.7	25.9
Body width	13.0	13.2	14.1	14.2	14.4	14.6	14.0	14.9
Snout length	4.9	6.0	4.9	6.2	5.2	5.4	4.7	5.4
Eye diameter	10.8	10.5	11.1	12.2	12.0	11.7	12.0	12.7
Interorbital width	9.2	9.1	8.7	8.5	8.3	8.7	11.1	9.5
Caudal peduncle depth	9.7	9.7	9.3	10.1	9.6	10.8	10.2	9.8
Caudal peduncle length	18.8	17.6	18.3	18.2	17.2	17.4	16.6	19.2
Pre-first dorsal distance	42.8	42.2	42.3	43.9	43.6	42.5	46.7	42.5
Pre-second dorsal distance	53.5	52.3	51.1	53.8	54.6	52.0	56.8	53.6
Preanal distance	51.8	53.0	50.8	55.2	53.4	52.6	54.6	52.6
Prepelvic distance	31.8	34.4	31.7	34.9	35.3	35.2	36.7	35.1
First dorsal-spine length	2.3	3.4	4.0	3.0	3.7	2.9	3.2	2.5
Longest dorsal ray	23.8	19.6	20.1	19.0	21.1	20.6	18.1	24.5
Anal-spine length	8.2	7.4	7.7	8.2	7.6	9.2	7.8	9.4
Longest anal ray	22.1	17.7	19.1	19.4	19.1	20.6	17.1	19.9
Pectoral-fin length	15.8	19.1	15.6	17.7	19.1	17.7	18.9	20.3
Pelvic-fin length	21.5	14.9	17.9	14.8	17.4	16.1	15.8	16.2
Caudal-fin length	25.4	24.2	24.4	27.4	21.2	20.1	27.9	27.1



Figure 11. *Kiunga filamentosa*, preserved holotype (WAM P.35286-001), 29.6 mm SL, tributary of Fly River, 8.4 km west of Drimkas Village, Papua New Guinea.

Photo: Gerald Allen

peduncle; fins semi-translucent grey or whitish with narrow black margin (missing on most specimens due to fin damage) on dorsal, anal, and caudal; also black stripe along bases of dorsal and anal fins, and diffuse blackish bar at base of caudal fin.

Comparisons: *Kiunga auromarginata*, n. sp. is easily distinguished from other members of the genus by a combination of features that include a slender body shape (depth 25.5–27.0, mean 26.0% of SL), relatively high number of dorsal and anal-fin rays (17–20 segmented elements; Table 1), 3–5 cheek scales, and a distinctive colour pattern that features bright orange margins and basal stripes on the dorsal and anal fins.

Distribution and habitat: This species is currently confirmed only from Taki Creek, a small tributary of the Elevala River, 4.3 km south of Gasuke Village (Fig. 1). In 2014 Horizon Oil also recorded *Kiunga* from two locations on Agime Creek, 20 km to the west-south-west of Taki Creek, and from an unnamed creek ~10 km to the east-north-east of Taki Creek. These records are assumed to be *K. auromarginata* given their proximity, however no specimens are available to confirm this identification. The habitat (Fig. 10) at Taki Creek consists of a lowland creek, approximately 5–8 m wide, 1–1.5 m deep, slow flowing, with clay banks and soft organic bed, some submerged macrophyte and woody debris, trailing riparian vegetation and dense rainforest canopy cover. Water was slightly turbid after overnight rains. Only small numbers were observed. Other species present at the site included *Craterocephalus nouhuysi*, *Oxyeleotris aruensis*, *Glossogobius concavifrons*, *Melanotaenia novaeguineae* (formerly identified as *M. goldiei*), and *Pseudomugil novaeguineae*.

Etymology: The species is named *auromarginata* (Latin: “gold margined”) with reference to the golden-orange fin margins.

Kiunga filamentosa n. sp.

(Figs. 1 & 11; Tables 2 & 4)

Holotype: WAM P.35286-001, 29.6 mm SL, small creek, tributary of Fly River, 8.4 km west of Drimkas Village, 5° 55.329'S, 141° 23.550'E, Papua New Guinea, seine net, A.W. Storey, 14 October 2018.

Paratypes: WAM P. 35286-002, 15 specimens, 18.2–28.4 mm SL, collected with holotype; WAM P.35287-001, 2 specimens, 21.8–22.5 mm SL, small creek, tributary of Fly River, 5.8 km west of Drimkas Village, 5° 54.939'S, 141° 25.012'E, Papua New Guinea, seine net, A.W. Storey, 14 October 2018.

Diagnosis: Dorsal-fin elements II–III + 15–18 (usually II + 15–16, mean 15.7); anal-fin elements I, 16–18, mean 17.1; second dorsal and anal fins relatively tall with triangular profile, longest dorsal ray 18.1–25.9, mean 21.6% SL and longest anal ray 18.2–35.9, mean 22.1% SL; first dorsal spine 1.7–3.6, mean 2.8% SL; anal spine relatively short, 6.3–10.7, mean 8.1% SL; undamaged pelvic-fin filaments of presumed adult male prolonged, extending well past anal-fin origin and sometimes to anterior portion of caudal peduncle when adpressed; lateral scales 28; transverse scales 5; cheek scales 3–4; predorsal scales 11–14 (usually 12–13), including pair of enlarged scales anteriorly; greatest body depth 27.8–31.4, mean 29.0% of SL; caudal fin forked; colour in life unknown, but presumably similar to other species with semitransparent body, narrow black midlateral stripe, and blackish margins on median fins; maximum known size 29.6 mm SL.

Description: Dorsal-fin elements II + 15 (II–III + 15–18); anal-fin elements I, 18 (I, 16–18); pectoral-fin rays 11 (11–12); pelvic-fin elements I, 5; branched caudal rays 10; principal caudal rays 12; accessory caudal rays 8 + 7 (6–8 + 6–8); lateral

scales 28; transverse scales 5; scales between dorsal fins 3; predorsal scales 13 (11–14); preopercle scales 4 (3–4, scale frequently missing and normal count probably 4); scales on opercular series 5; gill rakers on lower limb of first arch 7 (7–8).

Body depth 3.5 (3.2–3.6, 3.4); head length 4.5 (3.8–4.3, 4.1), pre-first-dorsal distance 2.4 (2.1–2.4, 2.2), pre-second-dorsal distance 1.9 (1.7–1.9, 1.8), preanal distance 2.0 (1.8–2.0, 1.9), prepelvic distance 3.2 (2.5–3.1, 2.8), all in SL. Greatest width of body 2.2 (1.8–2.3, 2.1) in body depth. Snout length 4.4 (3.7–5.2, 4.4); eye diameter 2.3 (2.0–2.4, 2.2); interorbital width 2.6 (2.6–2.8, 2.7); depth of caudal peduncle 2.3 (2.2–2.5, 2.3); length of caudal peduncle 1.1 (1.2–1.6, 1.3), all in HL.

Second dorsal fin originates slightly behind anal fin origin; longest spine of first dorsal fin 12.8 (7.1–13.7, 9.1), longest ray of second dorsal fin 1.0 (0.9–1.4, 1.2), of anal fin 1.0 (0.7–1.5, 1.2), all in HL. All segmented rays of second dorsal and anal fins branched. Pelvic fins variable in length, but most likely damaged in most specimens; four presumed male paratypes with filamentous rays extending well past anal-fin origin and in 26.6 mm SL specimen, elongate filament on right side extends to level of anterior caudal peduncle when adpressed, length of pelvic fin 4.4 (1.7–7.8, 4.8) in SL. Pectoral fins pointed, longest rays 1.4 (1.2–1.5, 1.4) in HL. Caudal fin forked, its length 3.7 (3.6–4.7, 4.1) in SL.

Jaws oblique, mouth terminal; maxilla extends posteriorly to level between anterior edge of eye and anterior edge of pupil. Upper jaw (premaxilla) covered with about 50 teeth arranged in 2 to 4 irregular rows on each side that are exposed when the mouth is closed, most teeth small and villiform with sharp tips, largest teeth situated along inner edge of rear portion of premaxillary, including 3–4 forward slanting teeth that are larger than others. Lower jaw with less conspicuous, small,

backward slanting villiform teeth confined mainly to anterior portion, about 15 to 20 on each side of median symphysis.

Scales cycloid, relatively large and elongated anteroventrally, arranged in regular horizontal rows; predorsal scales extending to middle of interorbital, including 2–3 relatively enlarged scales on frontal-interorbital; 4 (3–4) scales on cheek just below eye, those with 3 scales likely missing one scale; pores of cephalic sensory pores relatively large and conspicuous, 8 in dentary/preopercle series and 8 in supraorbital series; nostril openings relatively small, anterior naris placed on side of snout next to premaxillary and posterior naris situated next to upper anterior corner of eye.

Colour in life: unknown as photographs or colour notes of live or fresh specimens were not obtained.

Colour in alcohol (Fig. 11): overall yellowish white with pepper-like melanophores on scale margins on most of body except ventralmost portion and lower half of caudal peduncle; narrow black stripe along middle of side from pectoral region to base of caudal fin; concentration of melanophores on mid-dorsal line between origin of second dorsal fin and interorbital; weaker concentration of dark pigment on dorsal and ventral mid-line of caudal peduncle, abdomen, and pectoral-fin base; fins mainly semi-translucent grey except prominent black margin on dorsal and anal fins, and rear edge of caudal fin; less conspicuous black stripe along base of dorsal and anal fins, and blackish bar on caudal-fin base.

Comparisons: *Kiunga filamentosa*, n. sp. and *K. auromarginata*, n. sp. are generally separable from congeners by their higher number of segmented dorsal and anal-fin rays (Table 1) and presence of 3–4 cheek scales compared with usually 2 scales in the other species. *Kiunga filamentosa* is unique in possessing elongate, filamentous pelvic rays that extend well past the anal-fin origin, although this feature appears to be restricted to presumed males.

Table 4. Proportional measurements of holotype and paratypes (WAM.P.35286-001 and P.35286-002 respectively) of *Kiunga filamentosa* expressed as percentage of the standard length.

	Holotype				Paratypes					
Standard length	29.6	28.4	26.6	25.0	22.5	22.2	22.0	21.8	20.2	18.2
Head length	22.1	23.6	23.1	25.3	24.7	26.5	24.4	24.9	26.4	25.3
Body depth	28.4	28.8	30.0	29.4	28.4	28.2	30.0	31.4	28.5	30.0
Body width	12.9	12.4	12.9	14.0	13.8	14.6	14.4	13.8	14.3	14.5
Snout length	5.1	5.0	5.1	4.8	5.3	5.7	6.5	4.8	5.1	6.4
Eye diameter	9.7	10.4	9.8	10.8	11.7	12.0	11.9	11.2	11.9	12.8
Interorbital width	8.6	8.9	8.9	9.4	9.3	10.0	9.6	8.8	9.7	9.3
Caudal peduncle depth	9.6	10.1	10.2	10.6	10.5	10.8	11.1	11.4	10.9	11.6
Caudal peduncle length	20.6	18.3	17.9	19.2	18.6	17.1	17.5	20.6	17.4	20.3
Pre-first dorsal distance	42.4	43.4	44.5	43.8	43.8	46.2	45.6	45.9	46.8	47.1
Pre-second dorsal distance	51.8	53.3	53.8	54.4	52.1	55.0	57.4	56.8	59.3	58.7
Preanal distance	50.1	51.2	53.3	54.3	50.2	53.0	52.7	50.5	56.3	56.8
Prepelvic distance	31.0	32.2	32.3	35.0	34.5	38.3	36.0	35.1	39.3	40.2
First dorsal-spine length	1.7	2.9	2.5	3.0	3.3	2.8	3.4	3.6	1.9	2.7
Longest dorsal ray	21.8	27.1	24.5	23.2	25.9	21.4	18.5	21.0	18.9	20.6
Anal-spine length	8.2	9.8	8.4	8.7	8.9	7.9	7.3	7.8	7.0	9.6
Longest anal ray	21.7	33.9	27.0	24.0	24.6	18.2	18.7	20.8	18.9	20.2
Pectoral-fin length	15.3	18.3	16.9	17.1	18.5	19.3	19.7	17.4	19.3	20.3
Pelvic-fin length	22.8	41.2	59.0	19.4	41.4	17.7	19.9	37.3	15.3	17.1
Caudal-fin length	26.9	25.4	25.2	22.8	27.5	26.8	25.5	23.5	24.5	25.0

It further differs from *K. auromarginata* in having a noticeably deeper body (mean 29.0 versus 26.0% SL).

Distribution and habitat: The new species is presently known only from two sites, separated by a linear distance of 2.8 km, located along the newly constructed (2018) Drimkas Road at respective road distances of 6.7 and 9.8 km west of the Fly River (Fig. 1). The general area lies about 25 km northeast of Kiunga. Access to the creeks was only possible because of the newly constructed road, with sampling conducted around the road crossings, where the creeks emerged from dense tropical rainforest. These clear, slow flowing lowland creeks were approximately 5 m wide, 1–1.5 m deep, with a mostly soft organic bed in deeper pools interspersed by more shallow gravel runs.

Etymology: The species is named *filamentosa* (Latin: “filamentous”) with reference to the filamentous pelvic fins of males.

Kiunga leucozona n. sp.

(Figs. 1, 12–13; Tables 2 & 5)

Holotype: WAM P.34911-001, 33.0 mm SL, Ok Indok, tributary of Ok Tedi system, 5° 41.097'S, 141° 4.128'E, seine net, A. Harman *et al.*, 16 October 2014.

Paratypes: WAM P. 34911-002, 20 specimens, 17.3–33.3 mm SL, collected with holotype; WAM P.34918-001, 3 specimens, 21.1–28.4 mm SL, Ok Yep, tributary of Ok Tedi system, 5° 41.919'S, 141° 2.915'E, seine net, A. Harman *et al.*, 19 October 2014; WAM P.34919-001, 17.3 mm SL, Ok Yep, tributary of Ok Tedi system, 5° 41.199'S, 141° 2.040'E, seine net, A. Harman *et al.*, 19 October 2014; WAM P.34921-001, 13 specimens, 15.1–23.6 mm SL, Ok Yep, tributary of Ok Tedi system, 5° 42.405'S, 141° 1.424'E, seine net, A. Harman *et al.*, 20 October 2014; WAM P.34922-001, 3 specimens, 25.8–30.8 mm SL, Ok Birim,

tributary of Ok Tedi system, 5° 42.400'S, 141° 1.783'E, seine net, A. Harman *et al.*, 20 October 2014; WAM P.34923-001, 4 specimens, 23.0–26.2 mm SL, Ok Teelleck, tributary of Ok Tedi system, 5° 42.490'S, 141° 2.282'E, seine net, A. Harman *et al.*, 20 October 2014; WAM P.34924-001, 4 specimens, 19.9–26.6 mm SL, Ok Indok, tributary of Ok Tedi system, 5° 41.097'S, 141° 4.132'E, seine net, A.W. Storey, 1 March 2015; WAM P.34925-001, 4 specimens, 18.8–24.1 mm SL, Ok Barat, tributary of Ok Tedi system, 5° 41.518'S, 141° 3.680'E, seine net, A.W. Storey, 1 March 2015.

Diagnosis: Dorsal-fin elements I–III + 12–14 (usually II + 12–13, mean 12.7); anal-fin elements I, 14–17, mean 15.5; second dorsal and anal fins relatively tall with triangular profile, longest dorsal ray 21.1–34.5, mean 27.2% SL and longest anal ray 18.5–32.6, mean 26.7% SL; first dorsal spine 1.5–3.9, mean 2.7% SL; anal spine relatively short, 5.8–9.0, mean 7.5% SL; lateral scales 28; transverse scales 5; cheek scales 1–3 (usually 2); predorsal scales 13–14, including pair of enlarged scales anteriorly; greatest body depth 28.6–33.6, mean 31.7% of SL; caudal fin forked; colour in life mainly semitransparent with faint dark midlateral stripe and silvery peritoneum clearly evident; second dorsal, anal, and caudal fins with relatively broad black outer margin and adjacent submarginal white zone; outer edge of second dorsal and anal fins rarely with submarginal yellowish stripe at base of anterior rays; maximum known size 33.3 mm SL.

Description: Dorsal-fin elements III+13 (I–III + 12–14); anal-fin elements I, 16 (I, 14–17); pectoral-fin rays 12 (10–13); pelvic-fin elements rays I, 5; branched caudal rays 10; principal caudal rays 12; accessory caudal rays 7 + 7 (6–8 + 6–8); lateral scales 28; transverse scales 5; scales between dorsal fins 3; predorsal scales 14 (13–14); preopercle scales 3 (1–3); scales on opercular series 5; gill rakers on lower limb of first arch 8.

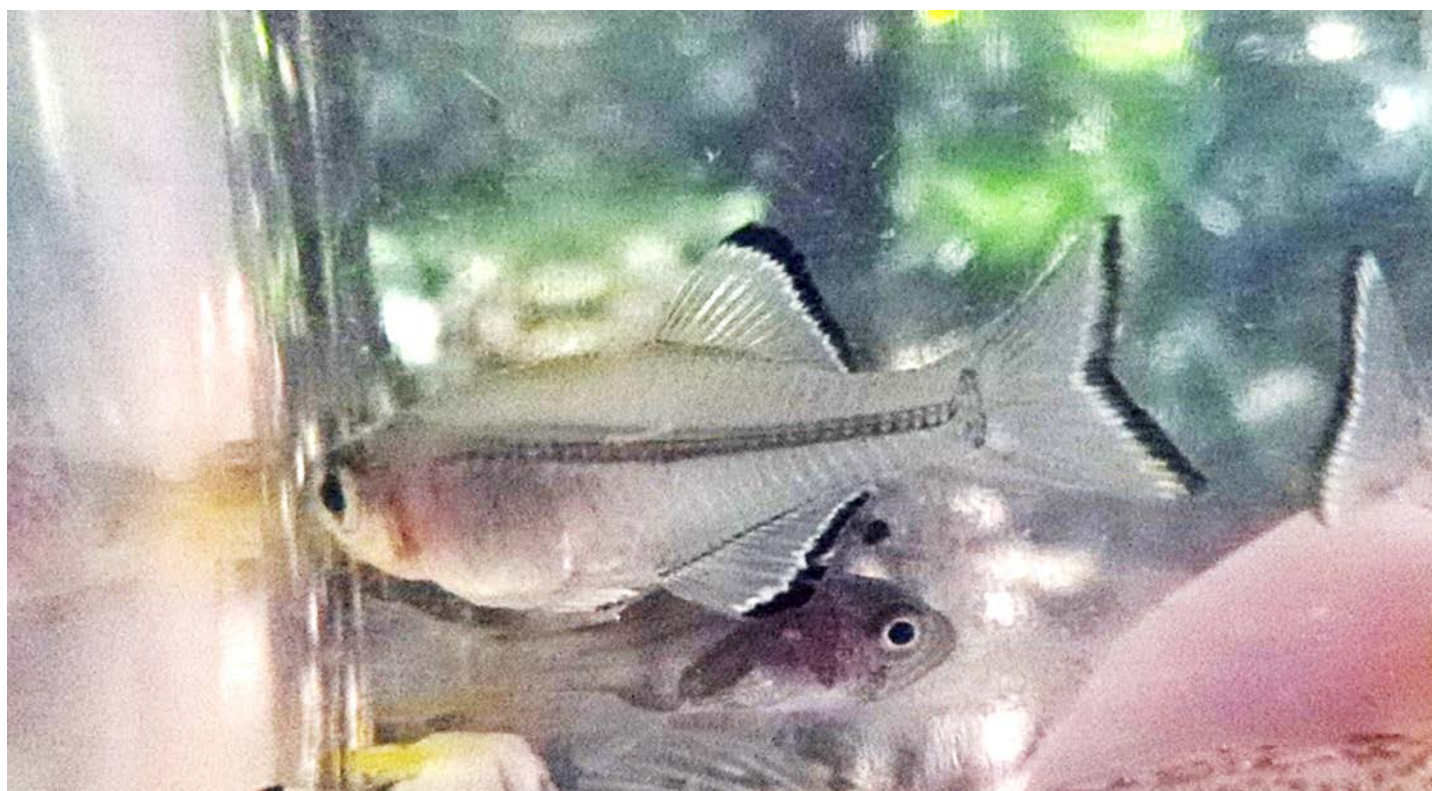


Figure 12. *Kiunga leucozona*, freshly collected specimens, approximately 30 mm SL, western tributary of Ok Tedi system, Papua New Guinea. Photo: Andrew Storey



Figure 13. *Kiunga leucozona*, preserved holotype (WAM P.34911-001), 33.0 mm SL, Ok Indok, tributary of Ok Tedi system, Papua New Guinea. Photo: Gerald Allen

Body depth 3.1 (3.0–3.5, 3.2); head length 4.5 (3.7–4.6, 4.13), pre-first-dorsal distance 2.2 (1.9–2.2, 2.1), pre-second-dorsal distance 1.8 (1.6–1.8, 1.7), preanal distance 1.9 (1.7–1.9, 1.8), prepelvic distance 2.8 (2.4–2.9, 2.7), all in SL. Greatest width of body 2.5 (1.9–2.6, 2.3) in body depth. Snout length 3.8 (3.3–4.7, 4.0); eye diameter 2.3 (2.0–2.3, 2.2); interorbital width 2.7 (2.4–2.7, 2.6); depth of caudal peduncle 2.2 (2.0–2.4, 2.2); length of caudal peduncle 1.2 (1.0–1.4, 1.2), all in HL.

Second dorsal fin originates slightly behind anal fin origin; first dorsal fin spine 8.9 (6.3–16.7, 9.2), longest ray of second dorsal fin 0.8 (0.7–1.2, 0.9), of anal fin 0.8 (0.7–1.3, 0.9), all in HL. All segmented rays of second dorsal and anal fins branched. Pelvic fin tips falling well short of anal-fin origin, pelvic length 1.6 (1.4–1.9, 1.7) in HL. Pectoral fins pointed, longest rays 1.3 (1.2–1.5, 1.3) in HL. Caudal fin forked, its length 3.5 (3.4–4.1, 3.7) in SL.

Jaws oblique, mouth terminal; maxilla extends posteriorly to level between anterior edge of eye and anterior edge of pupil. Upper jaw (premaxilla) covered with about 45–50 teeth arranged in 2 to 4 irregular rows on each side, including many that are exposed when the mouth is closed, most teeth small and villiform with sharp tips, largest teeth situated along inner edge of rear portion of premaxillary, including several forward-slanting teeth that are larger than others. Lower jaw with less conspicuous, small, backward slanting villiform teeth confined mainly to anterior portion, about 20–30 on each side of median symphysis.

Scales cycloid, relatively large and elongated anteroventrally, arranged in regular horizontal rows; predorsal scales extending to middle of interorbital, including pair of enlarged scales on frontal-interorbital; 3 (1–3) scales on cheek just below eye; pores of cephalic sensory pores relatively large and conspicuous, 7–8 in dentary/preopercle series and 9–10 in supraorbital series; nostril openings relatively small, anterior naris placed on side of snout next to premaxillary and posterior naris situated next to upper anterior corner of eye.

Colour in life (Fig. 12): mainly semitransparent with internal swim bladder clearly evident; lining of stomach silvery and opercle pinkish due to underlying gills; iris silvery; faint dark brown or blackish stripe along middle of side and concentrations of melanophores along bases of second dorsal and anal fins; fins generally semitransparent; second dorsal, anal, and caudal fins with relatively broad black outer margin and adjacent submarginal white band; outer edge of second dorsal fin rarely with submarginal yellowish stripe at base of anterior rays; pectoral and pelvic fins semitransparent, except pelvics sometimes yellow.

Colour in alcohol (Fig. 12): overall yellowish white with inconspicuous pepper-like melanophores on scale margins, mainly on upper half of body; single row of close-set melanophores forming midlateral black stripe extending from just above pectoral-fin base nearly to caudal-fin base; diffuse blackish bar across base of caudal fin; fins semi-translucent grey or whitish, most specimens with at least slight indication of black margins on dorsal, anal, and caudal, but sometimes missing due to abraded rays.

Comparisons: Meristic and genetic comparisons (see separate section at end of article) reveal *Kiunga leucozona*, n. sp., which occurs on the western side of the Ok Tedi, is most closely related to *K. ballochi* from eastern drainages. The closest populations of the two species are separated by approximately 21 km (Fig. 1). Although similar in general appearance the two species exhibit important modal differences for certain fin-ray counts. For example, 70% of *K. ballochi* specimens possess three dorsal-fin spines compared to only 18% for the new species, which usually has two spines (75%). Additionally, the new species usually has 11 pectoral rays (61.3% of specimens), compared to the more prevalent count of 12 rays (60% of specimens) in *K. ballochi*. Although we have obtained only a few photographs of freshly captured specimens, the new species appears to exhibit wider, more vivid black margins on the second dorsal, anal, and caudal fin compared with *K. ballochi*. It also lacks the extensive yellowish hue on the median fins and pelvic fins, which is typical



Figure 14. Ok Birim habitat of *Kiunga leucozona*. Photo: Andrew Storey

of *K. ballochi*, although a small amount of yellow is evident submarginally on the dorsal and anal fins, and at least one specimen of *K. leucozona* has yellow pelvic fins.

Distribution and habitat: *Kiunga leucozona* is presently known from a small area (approximately 23 km²) encompassing streams that form part of the lower Ok Birim catchment (Fig.

Table 5. Proportional measurements of type specimens of *Kiunga leucozona* expressed as percentage of the standard length. Asterisk (*) indicates damaged condition.

	Holotype	Paratypes								
		WAM P.34911	WAM P.31911	WAM P.34911	WAM P.31911	WAM P.34924	WAM P.34911	WAM P.34925	WAM P.34911	WAM P.34918
Standard length	33.0	33.3	32.2	28.7	27.6	26.6	25.1	24.1	23.2	21.1
Head length	22.0	21.7	22.0	22.0	21.8	23.0	24.0	26.8	22.7	26.1
Body depth	32.2	32.7	31.3	30.8	31.6	32.6	28.6	31.7	30.1	33.6
Body width	12.7	12.9	12.4	13.0	13.2	13.7	14.9	15.7	13.7	15.1
Snout length	5.8	6.0	5.6	5.9	5.8	6.1	7.2	6.7	6.1	5.6
Eye diameter	9.4	9.7	10.0	9.6	10.6	10.6	11.4	12.0	10.3	12.5
Interorbital width	8.3	8.3	8.2	8.3	8.4	8.4	10.1	10.4	8.6	10.7
Caudal peduncle depth	10.0	10.2	10.1	10.0	10.9	10.9	10.3	10.9	10.7	10.7
Caudal peduncle length	18.3	18.8	20.2	21.3	20.8	21.3	18.0	20.6	20.5	21.5
Pre-first dorsal distance	45.5	47.2	48.3	46.1	47.7	48.3	50.9	53.6	48.2	50.3
Pre-second dorsal distance	56.2	57.7	58.9	56.2	56.9	58.0	60.5	60.4	58.5	60.3
Preanal distance	53.5	54.1	55.2	51.9	54.9	53.7	56.9	58.5	55.0	58.5
Prepelvic distance	36.1	36.7	36.0	34.6	36.8	36.1	38.8	38.9	38.0	40.9
First dorsal-spine length	2.5	2.2	2.7	2.6	3.2	2.5	3.2	2.2	3.5	2.9
Longest dorsal ray	26.8	32.4	30.0	29.9	27.4	26.7	21.1	22.1	25.3	21.4
Anal-spine length	*	7.5	*	6.8	6.7	8.3	6.0	*	8.8	7.2
Longest anal ray	26.9	31.3	32.6	28.4	27.5	28.1	18.5	20.4	23.8	20.5
Pectoral-fin length	16.9	18.6	19.0	18.4	17.0	19.1	18.6	20.0	17.7	18.7
Pelvic-fin length	14.1	14.6	13.6	14.1	14.8	13.5	13.8	14.5	13.5	14.3
Caudal-fin length	28.3	28.4	27.6	25.2	29.1	24.6	27.7	25.6	27.6	27.2

1). The centre of this area lies approximately 10 km to the west and slightly south of the settlement of Ningerum, and to the west of the Ok Tedi main channel. *Kiunga leucozona* was sampled from eight of 21 creeks sampled in the Ok Birim area. Typically the creeks (Fig. 14) were clear, slow-flowing, with 100% canopy cover, some dappled sunlight, dense riparian and trailing vegetation, substantial large woody debris loads, predominantly organic substrate, with moderate presence of submerged macrophyte beds. Across the sites sampled, temperature was 24–27°C, pH 5.6–7.1, and very low conductivity of 4–19 µS/cm.

Etymology: The species is named *leucozona* (Latin: “white belt or zone”) with reference to the white submarginal zone on the median fins.

Genetic analysis

For the ‘phylo’ dataset with 33 individuals and 15,298 SNPs ML recovered one tree with a -ln score of -92266.212 (Fig. 15).

Support across all the deeper nodes of the tree was strong. The first branching lineage consisted of *K. bleheri*, sister to this lineage were two pairs of species, with *K. ballochi* and *K. leucozona* being sister species and *K. auromarginata* and *K. filamentosa* were the second pair of sister species. Based on the ‘PCoA’ dataset, the PCoA provided clear separation with *K. bleheri* separating on the first axis, while *K. ballochi* and *K. leucozona* separated on the second axis, along with a tight grouping of *K. auromarginata* and *K. filamentosa* reflecting their closer relationship (Fig. 16). The five species mostly have large numbers of fixed differences between them (Table 6). Most pairwise comparisons were between 847 and 1,905 fixed differences, while the two pairs of more closely related species had 637 fixed differences between *K. ballochi* and *K. leucozona*. The more closely related pair, *K. auromarginata* and *K. filamentosa*, had 183 fixed differences (Table 6).

For the Cytochrome *b* dataset with 20 individuals and 601 base pairs ML recovered one tree with a -ln score of -1253.405

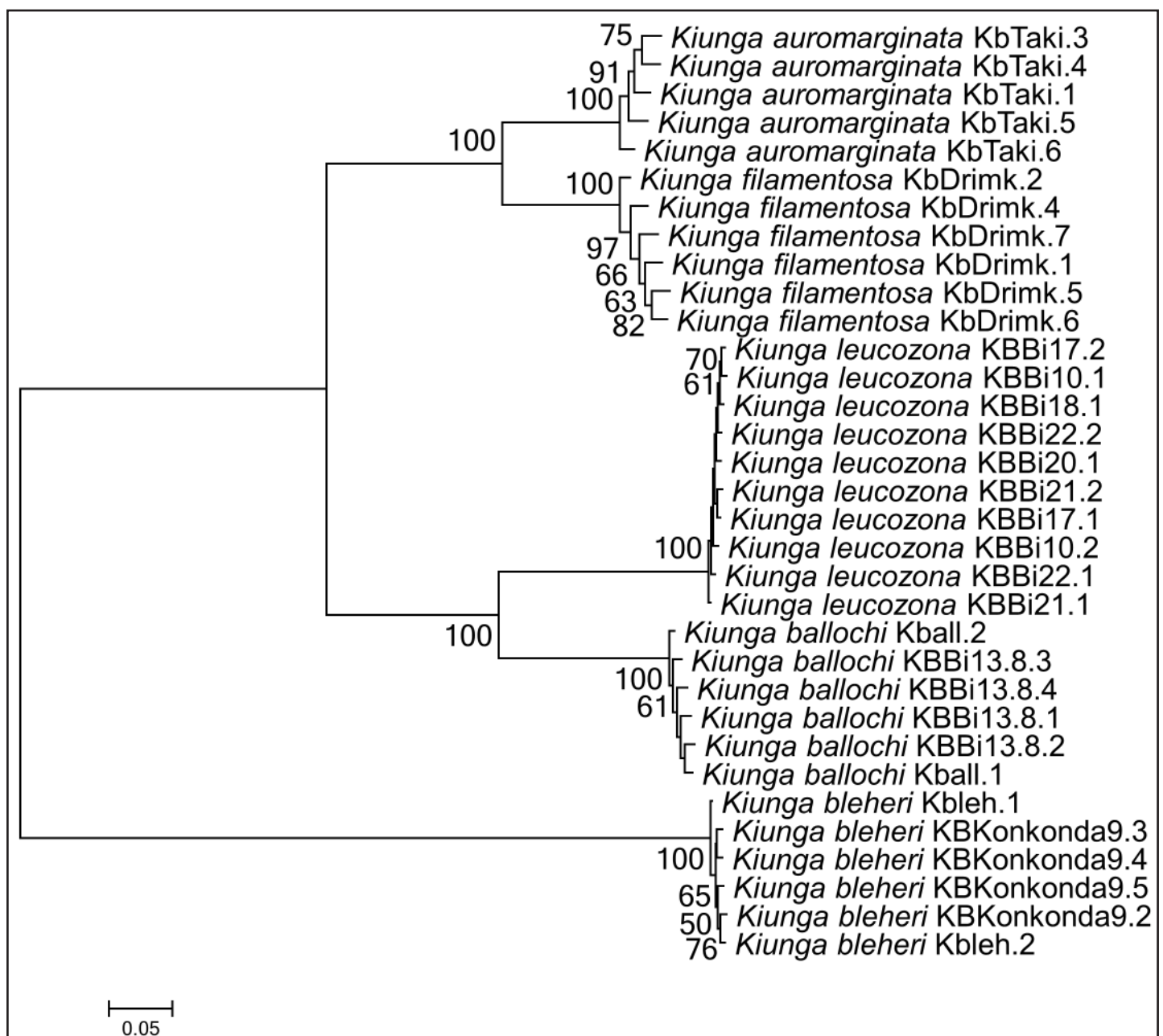


Figure 15. Maximum likelihood tree for five *Kiunga* species based on an analysis of 15,298 single nucleotide polymorphisms. Bootstrap values were obtained from 10,000 replicates. See Table 1 for locality details.

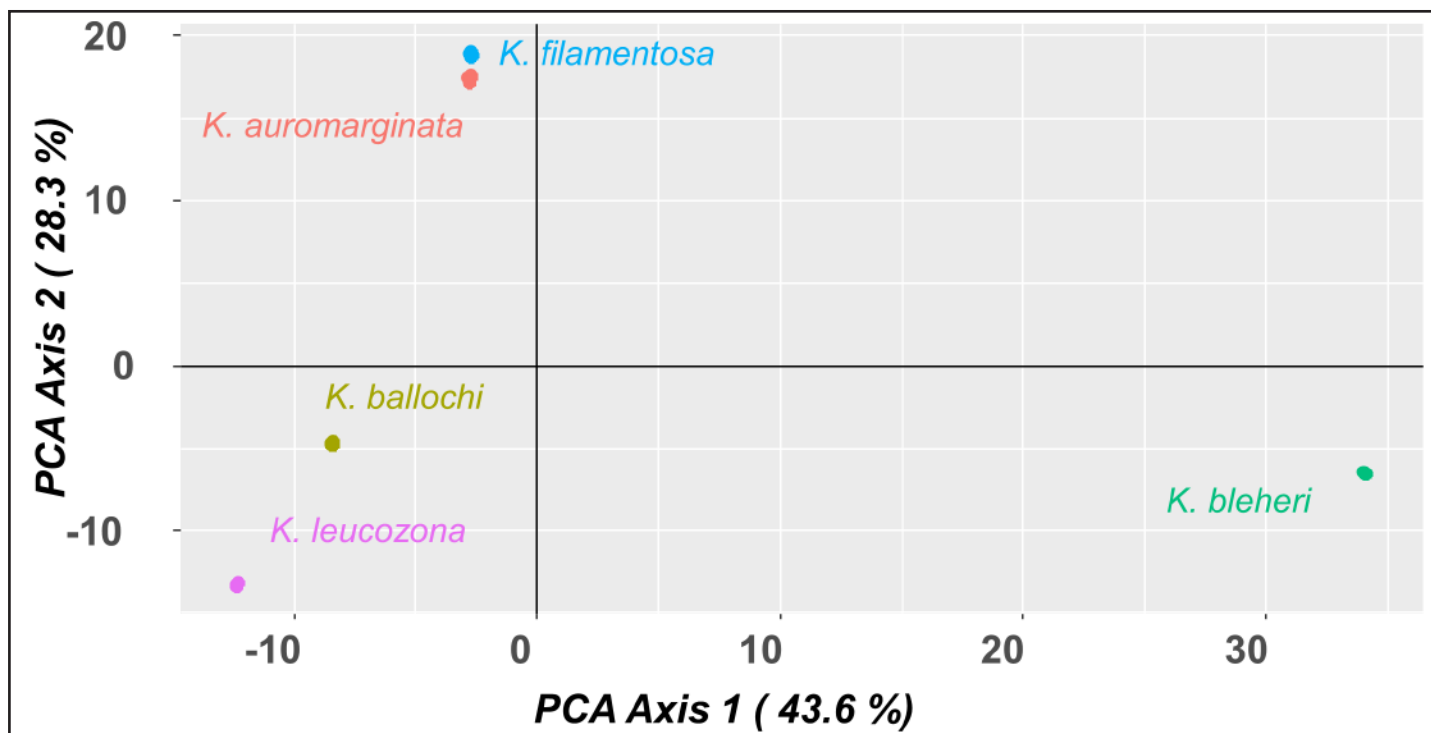


Figure 16. Principle coordinate analysis for five *Kiunga* species based on 4,469 single nucleotide polymorphisms.

(Fig. 17). Support across the two deeper nodes of the tree was weak, but each species was well resolved. The first branching lineage consisted of *K. bleheri*, the next was *K. ballochi* followed by *K. leucozona* and then the sister species of *K. auromarginata* and *K. filamentosa*. The differences in the phylogenetic branching patterns are only slightly different to the far more extensive SNP dataset and not surprising given the short fragment of mitochondrial DNA examined. Mean between group p-distances were all between 3.6 and 8.0% (Table 7). *Kiunga bleheri* consistently had high p-distances (6.2–7.9%) which reflect its position as the first branching lineage. The more closely related species pairs had values of 4.2% between *K. auromarginata* and *K. filamentosa* while *K. ballochi* and *K. leucozona* had a value of 3.9%.

Discussion

The genetic data successfully separate the five species, with even the closest pair of species, *K. auromarginata* and *K. filamentosa*, having high p-distances and fixed difference numbers. The levels of genetic divergence across these species given their close proximity to each other is unparalleled in any other monophyletic group of freshwater fishes in New Guinea or Australia. While it is difficult to accurately estimate the times of divergence between the species, even based on fast molecular clock rates for the Cytochrome *b* gene it is likely they diverged over the last 1–2 million years and they are clearly not recently evolved (e.g. in the last few hundred thousand years).

Each *Kiunga* species is separated by ~35 km of larger mainstream river channel, plus the distance upstream on side tributaries, which for most species is ~10–15 km, except *K. auromarginata* which occurs in tributaries of the Elevala River >50 km upstream from the Fly River (Fig. 1). So, most species are separated by ~70 km of river distance, with this value being in excess of 100 km for *K. auromarginata*. Although precise boundaries remain to be determined, this distribution pattern highlights the extraordinary close geographic proximity of the individual species in the genus. Based on the phylogeny (Figs. 15 & 17), *K. bleheri* is the most genetically distinct species and its range is situated near the junction of major rivers containing the other two upstream pairs of *Kiunga* species (Fig. 1). *Kiunga bleheri* is found in tributaries located in more downstream sections of the Ok Tedi system between the junction with the Fly River and upstream to the Ok Mart where it occurs at elevations of ~40–50 m. The northern portion of the range of *K. bleheri* is situated ~15 km south of *K. ballochi*, separated by the Ok Mart main channel (Fig. 1). The sister pair of *K. ballochi* and *K. leucozona* occur further upstream in the Ok Tedi system, separated by approximately 35 km of the main Ok Tedi channel, with *K. ballochi* found in tributaries to the Ok Mart just above its junction with the Ok Tedi at elevations of ~40–50 m. *Kiunga leucozona* occurs in elevations from ~65–90 m in tributaries to the Ok Birim, a subcatchment of the Ok Tedi. The more closely related sister pair of *K. auromarginata* and *K. filamentosa* occur further upstream in tributaries of the upper Fly River. *Kiunga auromarginata* occurs at an elevation of 80 m in the Elevala River system, which flows into the Fly River, while

Table 6. Number of fixed differences between five *Kiunga* species based on 4,469 SNPs.

	<i>K. auromarginata</i>	<i>K. ballochi</i>	<i>K. bleheri</i>	<i>K. filamentosa</i>
<i>K. ballochi</i>	867			
<i>K. bleheri</i>	1,740	1,905		
<i>K. filamentosa</i>	183	847	1,710	
<i>K. leucozona</i>	919	613	1,951	911

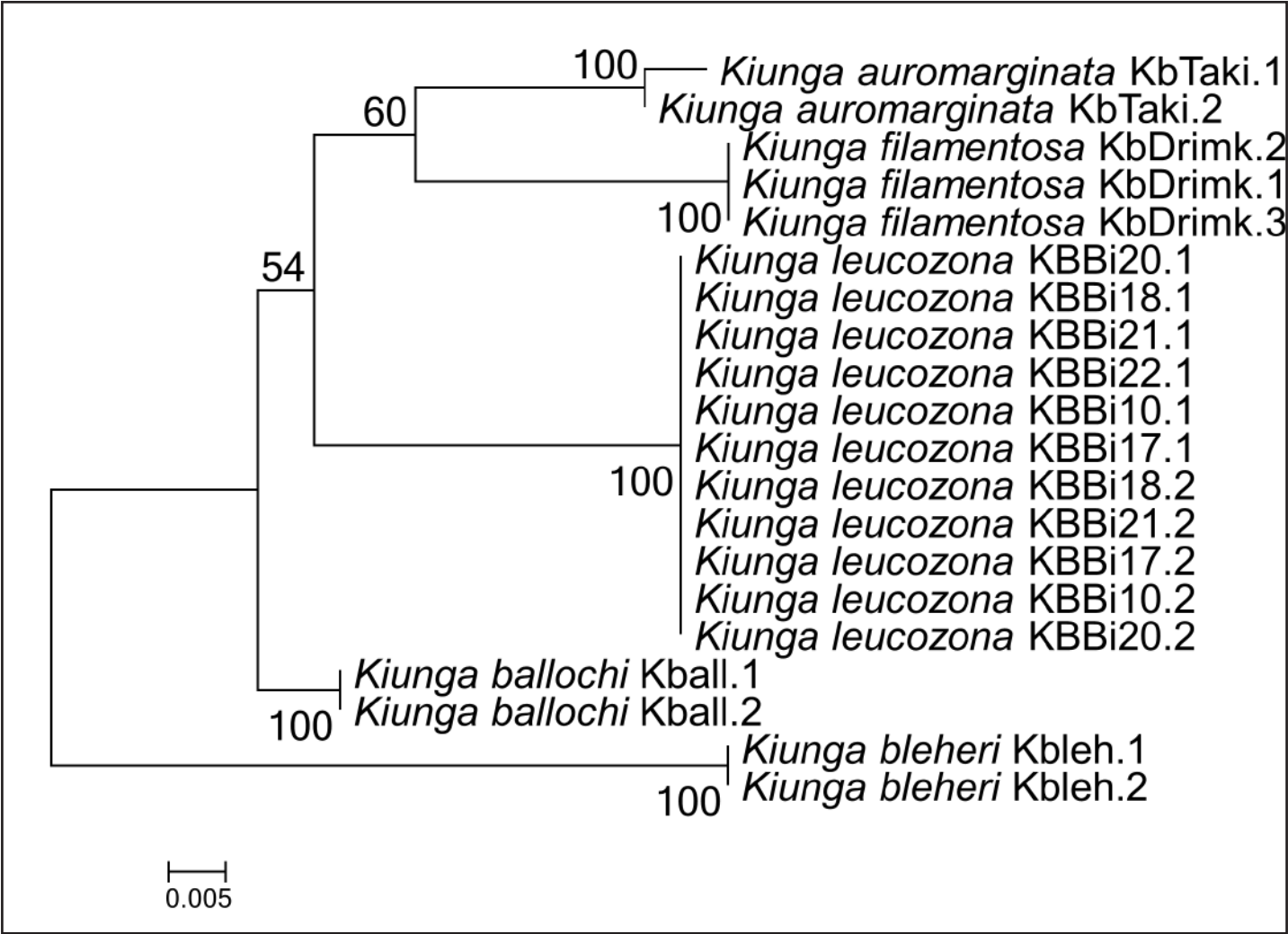


Figure 17. Maximum likelihood tree for five *Kiunga* species based on an analysis of 601 base pairs of the mitochondrial DNA gene Cytochrome *b*. Bootstrap values were obtained from 10,000 replicates. See Table 1 for locality details.

K. filamentosa occurs slightly further upstream at elevations of ~45–75 m in Fly River tributaries ~35–40 km upstream from the Elevala and Fly River junction. Despite the greater geographic distance between them, they have closer genetic divergences.

The patchy occurrence of recorded *Kiunga* species and the extensive still unsurveyed portions of this region, combined with the low abundance of *Kiunga* species, suggests there are likely additional species yet to be discovered in the Fly system, and possibly in catchments to the east and west. Their age of divergences (likely one to a few million years), and the occurrence of their sister group, two blue-eye species, *Pseudomugil connieae* and *P. furcatus* found ~800 km away to the east on the biogeographically distinct north-eastern side of New Guinea, suggests there has been extensive historic extinction of lineages within this group. However, they are clearly able to persist in small pockets of habitat for long time periods. Given the general lack of clear biogeographic barriers across much of southern New Guinea, based on the widespread occurrence of

many species (albeit likely with considerable phylogeographic structure) or closely related species, there is no reason to expect that *Kiunga* species could not be found further to the west in Papuan Indonesia.

The five *Kiunga* species are highly threatened with extinction due to pressure from substantial increases in human populations in the region, sedimentation impacts from unpaved roads, expanding populations of exotic fishes, potential land use changes and accidental and intentional damage to mining infrastructure and pipelines leading to spills of copper and pyrite concentrate (Amick et al., 2020). In early collections when the first major roads provided access, *Kiunga* individuals seemed common, as was the case for the more recent collection of *K. filamentosa* on the newly constructed road to Drimkas (Fig. 1). Increasing human settlements and impacts from the unpaved roads have led to the significant decrease or disappearance of some populations (Amick et al., 2020). Another blue-eye species, *Pseudomugil paskai* which was described from near

Table 7. Mean percentage between group p-distances between the five *Kiunga* species based on 601 base pairs of Cytochrome *b*.

	<i>K. auromarginata</i>	<i>K. ballochi</i>	<i>K. bleheri</i>	<i>K. filamentosa</i>
<i>K. ballochi</i>	3.6			
<i>K. bleheri</i>	7.9	6.2		
<i>K. filamentosa</i>	4.2	4.1	7.1	
<i>K. leucozona</i>	4.7	3.9	8.0	6.2

Kiunga (very close to the type locality of *K. bleheri*) has not been found in the area (or anywhere else in the Fly system) for at least the last 30 years despite efforts to find them. Hopefully these species can persist in more remote sections of these catchments where human populations and other impacts are far less (i.e. protection of primary rainforest is a key conservation priority), although exotic fishes will no doubt gradually spread and colonise many of these habitats. Since the first records of an introduced fish species in the Fly River in 1988, the Climbing Perch, *Anabas testudineus* (Anabantidae) (Storey et al. 2002), there are now an additional four species commonly encountered in floodplain habitats; Snakehead *Channa striata* (Channidae), Walking Catfish *Clarias batrachus* (Clariidae), Tilapia *Oreochromis mossambicus / niloticus* (Cichlidae), and Common Carp *Cyprinus carpio* (Cypridae), with Common Snowtrout *Schizothorax richardsonii* (Cyprinidae) common in the upper Strickland system, and the Himalayan or Golden Mahseer *Tor putitora* (Cyprinidae) identified from adjacent Gulf rivers (A.W. Storey, unpub. dat.). Thus, there is some urgency to locate and conserve any extant populations that may eventually be threatened.

Acknowledgements

We acknowledge the traditional owners of the Fly River region, and recognize their strong cultural connection to land, water and fishes.

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